WEEK 9 OF 15 · STAGE 2: GCAM IN DEPTH

Land, water and bioenergy

Why a climate policy ends up on the farm

New R skill rmap for maps; first look inside gcamdata: how R turns source data into GCAM input files

Modelling Climate and Energy Futures · beginner-friendly lecture and lab slides

COURSE MAP

Where we are in the course

Stage 1 **Foundations**	Week 1 Why model the future?	Week 2 Climate primer	Week 3 Energy primer	Week 4 Economics primer	Week 5 Cost-benefit IAMs: DICE	Week 6 The scenario framework
Stage 2 **GCAM in depth**	Week 7 Inside GCAM	Week 8 Energy in GCAM	**Week 9** **Land, water and bioenergy**	Week 10 Emissions and policy		
Stage 3 **Advanced methods**	Week 11 Optimisation IAMs	Week 12 Why models disagree	Week 13 Uncertainty	Week 14 Frontiers and limits	Week 15 Projects and review	

TODAY'S GOALS

By the end of today you can

1. Calculate profit per hectare and land shares for competing uses
2. Estimate the land needed for a given amount of bioenergy
3. Explain why water is balanced by river basin, not by region
4. Draw maps of GCAM results with rmap and find where an input file comes from in gcamdata

? **Big question** If we grow crops for energy, where does the land and water come from, and what gives way?

A carbon price reaches the countryside

Put a price on carbon for power plants. Soon farmland gets more valuable, forests may grow, and some rivers start to run short.



How can a policy aimed at power stations change what farmers plant?

AN EVERYDAY PICTURE

Land is rented to the highest bidders



Imagine every patch of land is a flat for rent. A wheat farmer, a forester and an energy-crop grower all bid. Each can pay up to what the land earns them. The best payers win more flats, but never all of them, because soils, climates and habits differ.

What it means for models

1. The bid is profit per hectare
2. Higher profit wins more land, gradually
3. Protected land is not for rent

Worked example: wheat or energy grass?

Problem Profit per hectare = (price − cost) × yield. Wheat: $200/t, cost $120/t, 3 t/ha. Grass: $60/t, cost $30/t, 10 t/ha.

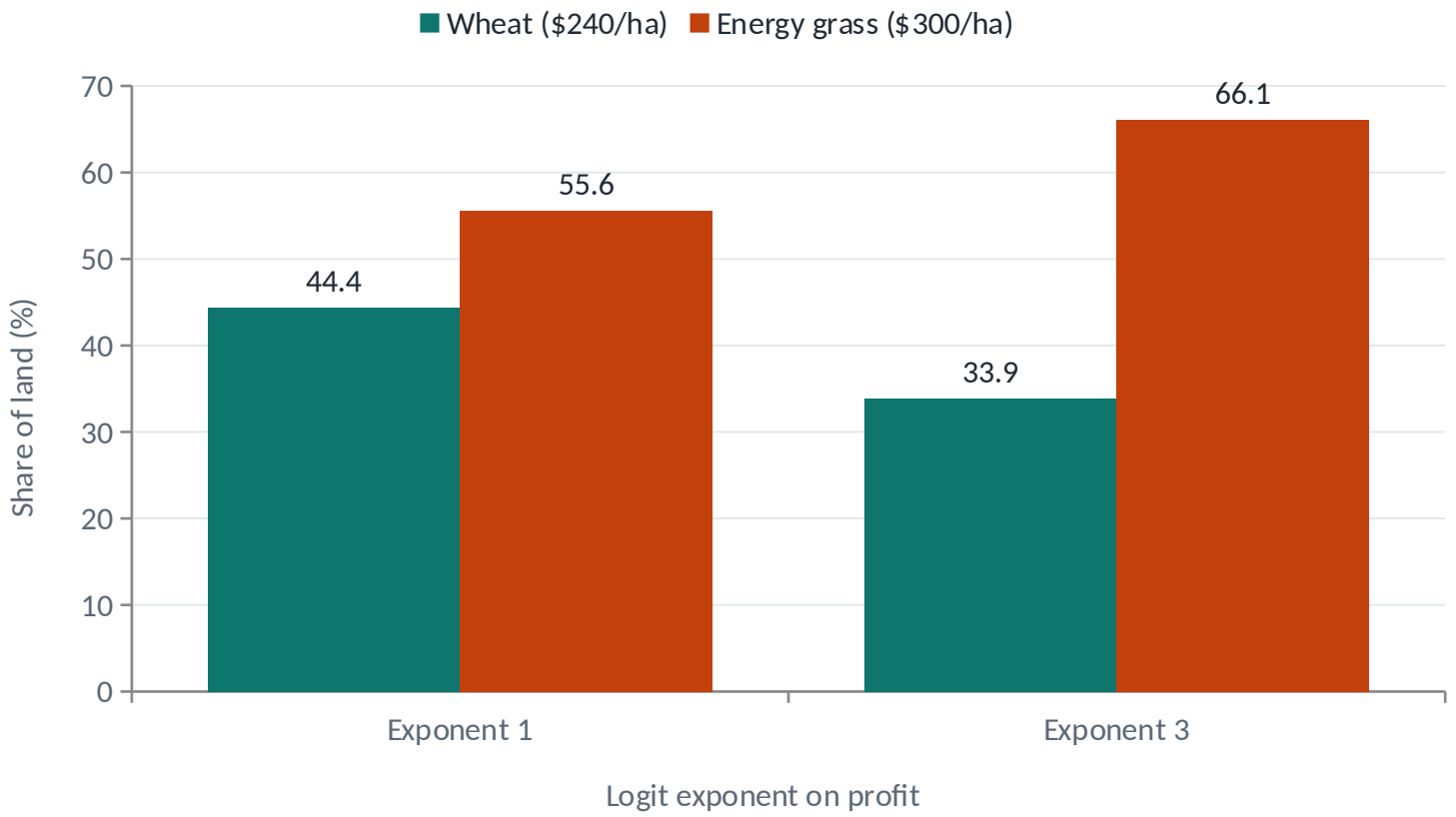
1. Wheat: (200 − 120) × 3 = 80 × 3 = $240 per ha
2. Grass: (60 − 30) × 10 = 30 × 10 = $300 per ha
3. Share by profit, exponent 1: total = 240 + 300 = 540
4. Wheat: 240 ÷ 540 = 44%; grass: 300 ÷ 540 = 56%

Answer

Wheat 44%, energy grass 56% of the contested land

PICTURE THE NUMBERS

A stronger exponent moves more land to the winner



Illustrative, computed from the share rule

Read it like this

1. Exponent 3: 240^3 = 13.8 million, 300^3 = 27.0 million
2. Shares 34% and 66%
3. A carbon price raises grass profit, so it gains more

Quick check

Maize: $180/t, cost $100/t, yield 4 t/ha. Grass earns $300/ha. With exponent 1, what share goes to maize?

- **A** About 37%
- **B** About 48%
- **C** About 52%
- **D** About 60%

Quick check: answer

Maize: $180/t, cost $100/t, yield 4 t/ha. Grass earns $300/ha. With exponent 1, what share goes to maize?

A About 37%

B About 48%

C **About 52%** ✓

D About 60%

HOW IT WORKS

Land is shared in nested rounds

1

All land

Protected land is set aside first and stays put

2

Big uses

Cropland, forest, pasture and shrubland compete

3

Crops

Wheat, maize, rice, energy crops and more compete

4

Water use

Each crop splits into irrigated and rainfed

Land is solved in more than 300 land subregions. Energy crops appear only in future periods.

Worked example: land for 10 EJ of bioenergy

Problem How much land does 10 EJ a year of energy grass need? Yield 10 t/ha; energy content 18 GJ per tonne.

1. Energy per hectare: 10 t × 18 GJ = 180 GJ per ha
2. 10 EJ = 10 × 10^9 GJ
3. Land = 10 × 10^9 ÷ 180 = 55.6 million ha
4. Halve the yield to 5 t/ha: land doubles to 111 million ha

Answer

About 56 million hectares at 10 t/ha, twice that at 5 t/ha

SIDE BY SIDE

Each river basin has a staircase of water supply

Source	Cost ($ per m^3)	Available
Renewable water (runoff)	0.05	first 100 units
Groundwater (finite stock)	0.20	next 15 units
Desalination	1.00	anything beyond 115

Demand 110 → price $0.20. Demand 125 → price $1.00. Illustrative numbers.

COMPARE

Why the map matters for water

Region view

- One number for the whole region
- Can look comfortable on average
- Hides where the stress is

Basin view

- One market per river basin
- A basin can cross borders
- Shows the one dry basin carrying most withdrawals

KEY IDEA

The nexus: follow one carbon price

1

Energy → land

A carbon price raises demand for bioenergy, so energy crops become more profitable.

2

Land shifts

Energy crops take land from food, pasture or forest, and crop prices rise.

3

Land → water

More irrigated crops raise withdrawals in some basins and push water prices up.

4

Water → energy

Power plants need cooling water, and fertiliser and farm fuel need energy.

WATCH OUT

Common mix-ups

Not quite	Better
The region has plenty of water, so no water problem	Water clears by basin. One basin inside the region can be badly stressed.
Bioenergy is free carbon removal	It uses land that could grow food or forest, plus water and fertiliser.
The most profitable crop takes all the land	Land shares follow a logit rule, so shares shift gradually.

Words to add to the class glossary

Yield How much a crop produces per unit of land, for example 3 tonnes of wheat per hectare.

Profit per hectare (Price − cost) × yield. The amount a land use earns on one hectare; GCAM shares land using it.

River basin The area drained by one river system. GCAM balances water supply and demand basin by basin.

Water withdrawal Water taken from a river, lake or aquifer. Part is returned; the part not returned is called consumption.

Nexus The links between energy, water and land: a change in one system moves the other two.

HANDS-ON LAB

Lab 9: maps and a first look in gcamdata

Map cropland by region and water withdrawals by basin with rmap, then open gcamdata and find which file holds solar's capacity factor.

Script: labs/lab9.R

4 steps

Stuck? Read the error aloud, check the slide, then ask

Map cropland by region

Add up cropland per region in 2050, then match GCAM's region names to the map's names.

```
prj <- loadProject("output/gcam_course.dat")
crops <- prj |>
  getQuery("aggregated land allocation") |>
  filter(scenario == "Reference", year == 2050,
         landleaf == "crops") |>
  group_by(region) |>
  summarise(value = sum(value)) |>
  mutate(subRegion =
           str_replace_all(region, "-", "_"))
rmap::map(crops, shape = rmap::mapGCAMReg32,
          save = FALSE)
```

Map water by river basin (gap 1)

Turn GCAM's water resource names into basin names, then draw them on rmap's basin map. Fill the gap with the basin map's name.

```
basin <- prj |>
  getQuery("water withdrawals by basin") |>
  filter(scenario == "Reference", year == 2050) |>
  mutate(subRegion = str_remove(
    basin, "_water withdrawals")) |>
  group_by(subRegion) |>
  summarise(value = sum(value)) |>
  left_join(rmap::mapping_gcambasins,
            by = "subRegion") |>
  select(subRegion = subRegionMap, value)
rmap::map(basin, shape = rmap::___,  # basin map
          save = FALSE)
```

Open gcamdata (read only)

gcamdata is the R package that builds every GCAM input file. It is a chain of chunks; each lists what it reads and writes.

```
gcamdata_dir <- path.expand("~/gcamdata")
devtools::load_all(gcamdata_dir)
chunks <- find_chunks()
count(chunks, module)
inputs_of("module_energy_L223.electricity")
outputs_of(
  "module_energy_elec_tech_scenarios_xml")
```

⌨ *Type it rather than paste it, run one line at a time (Ctrl+Enter), and check the output before the next line.*

LAB · STEP 4 OF 4

Find solar's capacity factor (gap 2)

Read one assumption file. Lines starting with # are comments. Fill the gap with the subsector that holds the PV technology.

```
cf <- read_csv(file.path(gcamdata_dir,
  "inst/extdata/energy",
  "A23.globaltech_capacity_factor.csv"),
  comment = "#")
filter(cf, subsector == "___")  # holds PV
```

Fill each ___ gap before running that line. Stuck? The solution file has the complete version.

Did it work?

You should see

- A world map in 32 pieces with no unmatched-region warning
- A basin map where a few basins stand out dark
- filter(cf, ...) shows the PV row with values by year

Worksheet questions

1. Which three regions have the most cropland in 2050 under Reference?
2. Pick a basin that is dark on your map. Why might a region total hide its stress?
3. Which CSV file holds solar's capacity factor, and which chunk reads it?

RECAP

Take-home messages

1. Land goes to uses with higher profit per hectare, shared gradually by a logit rule
2. Bioenergy land = energy wanted ÷ (yield × energy per tonne); low yields need much more land
3. Water markets clear by river basin, priced by the most expensive source needed
4. Energy, land and water are solved together: the nexus

Next: week 10

Week 10 adds emissions and policy: carbon prices, net-zero limits and carbon removal, and your first change inside gcamdata.

Practise

Interactive course, week 9: try the land explorer and the 5-question quiz.

Readings: rmap package guide; gcamdata overview; GCAM v8.2 documentation: Land, Water Supply, Water Demand, GCAM Data System